

AI Is Here To Stay: Misinformation and Human-Centric Models Between Risks and Opportunities

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For more than half a century, artificial intelligence remained a matter of speculation for novelists and philosophers. This changed in 2017 due to the introduction of the transformer architecture, spearheading the AI boom whose effects are still scarcely understood. This paper explores potential risks and use cases of this technology, with a "human-centric" cut: the first half showcases generation and detection of fake news and deepfakes, as well as how much LLMs can influence people's opinions and beliefs. The focus then shifts to the AIs themselves, about their own biases and their susceptibility to external stimuli, concluding with a quick overview on the state of research about AIs for positive social impact.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Artificial intelligence**; **Natural language processing**; **Reasoning about belief and knowledge**; Cognitive science.

Additional Key Words and Phrases: AI, NLP, Misinformation, Deepfakes, Opinions, Biases, AI4SI

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1 INTRODUCTION

Politicians are some of the most influential people in human society: from the Greeks' demagogues to our Presidents and Prime Ministers, leaders have always tried to steer society. Shapiro and Page (1984) [23] showed how nearly all 20th century U.S. Presidents have tried to lead public opinion, with varying degrees of success. As societies became more democratic, so grew the leaders' responsibility to convince potential followers that their policies are beneficial. Thus, speeches play a vital role in the functioning of society [2]: leaders depend on the verbal power to persuade people [5].

Moreover, democratic stability depends on citizens on the losing side accepting election outcomes. Clayton et al. [7] evaluated the effects of exposure to multiple statements from (at the time) former president Donald Trump attacking the legitimacy of the 2020 US presidential election. They found no evidence indicating that elites can erode democratic norms easily or that the effects of norm violations are uniform across the entire population. However, elite rhetoric can shape normative beliefs in core democratic values such as confidence in elections and support for peaceful transfers of power. This was especially evident amongst people who approved of Trump's performance in office.

Having established the importance of speeches, we can differentiate between two types: *internal speeches*, aimed at other institutional bodies, such as parliamentary speeches, and *external speeches*, aimed at the population, such as public speeches, interviews and social network posts. This last category is especially interesting since it affords politicians a direct and high-volume communication channel to their potential followers: U.S. President Donald Trump tweeted

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57,000 times between May 2009 and January 2021 alone [19], while Italy’s Prime Minister Giorgia Meloni totalled more than eighty million social network interactions in 2025 [27].

On the other hand, internal speeches are especially easy to access, being already aggregated in public databases, yet their technical and codified nature makes their interpretation harder for both human and automated agents. Moreover, until 1985 Italian stenographic documentation did not report speeches word for word, but rather translated them into a “language of the parliament” [14, 21].

2 SOCIAL NETWORKS

Broadly speaking, the goal of this project is to gather a corpus of politicians’ speeches and perform linguistic analyses on it. The idea is to start from standard NLP tasks such as word frequency and text complexity, to then move to classification tasks such as hate speech recognition, and finally to fine-tuning an LLM.

This objective is why I ultimately decided to gather a corpus of *external speeches* rather than internal ones. Large language models are trained with tokenizers, and the resulting token distribution is highly imbalanced: Chung et al. [6] did a controlled study that scaled the vocabulary of the language model from 24K to 196K while holding data, computation, and optimization unchanged. They discovered that models are disproportionately optimized on the part of language that appears most often, meaning they are way better at understanding common words and patterns. This translates to the fact that internal speeches are less understandable for an AI agent.

As previously stated, external speeches include social network communications. While television is still the first source of information for the majority of the population, online platforms are steadily growing and are already the preferred media for young people [12, 20]. There is evidence of an increase in political participation due to social media usage, as well as risks for the functioning of democracy [1, 18], and there have been attempts to predict election results with social media data analyses. Rita et al. measure sentiment polarity on Twitter, and conclude that tweets’ sentiment is not a reliable election results predictor. Additionally, results also show that it is impossible to state that social media impacts voting decisions [25].

Opposite results can be seen in Belcastro et al.’s [3] 2022 study: a real-time analysis was carried out during the 2020 US presidential election campaign, correctly identifying the leading candidate before Election Day in 10 out of 11 swing states.

Silva et al. [26] managed to match tweets against parliamentary speeches to measure politicians’ sentiment on a specific topic. They find that parliament members who participate less in parliamentary debate tend to have larger differences with their party on Twitter, suggesting that a certain level of self-censoring is taking place in the parliamentary arena.

Lastly, social networks open the possibility of semi-automatized corpora building.

3 PROJECT GOAL

Let’s now define the project’s goal.

I want to compare politicians’ speeches among three countries, namely Italy, France and the United States, between 1945 and 2025, and then cross-compare the resulting analyses with democracy level indicators taken from the V-Dem dataset¹ and fine-tune an LLM to better interpret political speech and classify populism and hate speech.

¹The V-Dem Dataset is a multidimensional and disaggregated dataset that reflects the complexity of the concept of democracy as a system of rule that goes beyond the simple presence of elections: <https://www.v-dem.net/data/the-v-dem-dataset/>.

3.1 Comparing Italy, France and the U.S.

Italy was a natural choice being the country I was born and raised in, therefore the one I am most knowledgeable about, but more than that, it has always been a fascinating research case study, being often referred to as an "anomaly" or, by authors such as Marc Lazar, as a kind of *laboratory* of a general change of European democracies. Since the 1990s, the state of Italian democracy has raised numerous questions, stimulating the curiosity of the international media, causing heated debates, providing material for seminars and conferences, and engendering academic and journalistic publications by Italians and foreigners alike. Lazar's opinion is that Italy is "at the same time an indicator of a more general democratic malaise and of the multiple attempts of invention of a new democratic order that are emerging all over the European Union". Similar phenomena affect, in their own ways, other European democracies, making it a meaningful case study [17].

Italian, French, and American histories intersected often during the past century: being at times allies, at times enemies during the Second World War, in the ensuing Cold War the United States encountered unexpected challenges from the Italian and French Communist Parties, the two strongest and most anti-American in Western Europe. Brogi [4] argues that the resistance to Americanization was necessary to legitimize the French and Italian Communist Parties' own existence.

Moreover, each country illustrates a different approach to democratic governance. They use three different executive models: the U.S. opted for a full presidential, where the President holds all power, France uses a semi-presidential system where power is split between the President and the Prime Minister, and Italy a Parliamentary one where the Prime Minister holds the power but depends on the Parliament. The party system is also different, since France and Italy have multi-party systems while the U.S. is strongly bipartisan, being divided between Democrats and Republicans. Lastly, the state system employed is also quite different: France's state is unitary and vertical, while Italy's regions hold strong power and independence. The U.S., on the other hand, is a Federation.

3.2 Roadmap

The first step is building corpora of politicians' speeches, one corpus for Italian speeches, one for French speeches and one for American speeches. To limit the scope, I decided to take only speeches from Italian Prime Ministers, U.S. Presidents, and either French Presidents when the President and the Prime Minister come from the same party, or French Prime Ministers when the President and Prime Minister are in opposition to each other. This is justified by the fact that when they are politically aligned, the President dominates. During cohabitation (opposing political camps), the Prime Minister gains real policy autonomy [22].

Speeches can be taken from different sources, such as official speeches, interviews, or social network posts, and can be in different media formats, such as textual, audio or video. For the latter two, a translating tool will be needed (such as a speech-to-text library).

The second step is using the corpora to perform linguistic analyses. The corpora can be subjected to data augmentation², since we can expect the density of speeches to decrease as we go back in time. Then, the corpora will be pre-processed, with procedures such as tokenization, stemming and lemmatization. We can then compute word frequency and text complexity measures, as well as hate speech and populism identification.

²In its most general sense, data augmentation denotes methods for supplementing so-called incomplete datasets by providing missing data points in order to increase the dataset's analyzability [15].

The third step is cross-referencing the linguistic metrics with democratic indicators such as freedom of expression, freedom of association, transparency and fairness of law and other such metrics. All democracy metrics can be found in the V-Dem dataset.

The fourth step is to fine-tune a pre-trained transformer model, for example Ministral 8B³, to make it better understand political speeches in three different languages, and to train it in a classification task for hate speech and populism recognition.

4 BUILDING THE CORPORA

The natural first step was to look at other corpora that aggregate similar data.

Europarl dataset [16] aggregates proceedings of the European Parliament in twenty one different European languages⁴ with the goal of generating sentence aligned text for statistical machine learning translation systems. The overall structure is akin to a collection of plaintext with minimal tagging.

ItaParlCorpus [9], on the other hand, is a comprehensive, annotated, and machine-readable database of Italy's parliamentary speeches spanning from 1948 to 2022. Each speech is identified by a precise date, the role of the speaker and its party name and family, and by the legislature number. It is not, whoever, very rich in terms of linguistic processing tagging.

IMPAQTS [8] is a multimodal corpus of around 2.65 million tokens including 1,500 speeches uttered by 150 prominent politicians spanning from 1946 to 2023. For each speaker, the corpus contains 4 parliamentary speeches, 2 rallies, 1 party assembly, and 3 statements (in person or broadcasted). The goal is to annotate political rhetoric that hides a deeper meaning, for example by employing sarcasm. There are four main types of annotation: implication (conventional, generalized, etc), presupposition (e.g., pragmatic), vagueness (syntactic, semantic, metaphoric), and topicalization (either syntactic or prosodic).

Parola di Leader, and specifically the corpus LP4 [13], is a corpus of more than five million occurrences taken from the stenographic archives of the Chamber of Deputies of Italy⁵, between 1948 (first legislature of the Italian Republic) and 2011 (excluding Monti Government, the 61st legislature).

They selected a list of thirty "highly influential politicians"⁶ and then for each of them they selected some speeches. Overall, they collected 1877 distinct speeches.

The corpus could be considered a corpora made of thirty distinct corpus, one for each politician. Each of these has, among others, the following features: name of the politician, role of politician (for example Prime Minister), whether the politician is the major party or in the opposition, number of legislature and number of seating of the day, and date of the seating, argument of the speech, name of the current Government (e.g., *Berlusconi I*), and lastly a reference to the file that holds the actual speech.

³Ministral 8B is a powerful edge model with extremely high performance/price ratio: <https://docs.mistral.ai/models/model-cards/ministral-8b-24-1>.

⁴All 21 Europarl languages: Romanic (French, Italian, Spanish, Portuguese, Romanian), Germanic (English, Dutch, German, Danish, Swedish), Slavik (Bulgarian, Czech, Polish, Slovak, Slovene), Finni-Ugric (Finnish, Hungarian, Estonian), Baltic (Latvian, Lithuanian), and Greek.

⁵Chamber of Deputies official online public archive: <https://legislatureprecedenti.camera.it/>.

⁶The full list of politicians follows: Giorgio Almirante, Giuliano Amato, Giulio Andreotti, Enrico Berlinguer, Silvio Berlusconi, Pier Luigi Bersani, Fausto Bertinotti, Rosy Bindi, Emma Bonino, Umberto Bossi, Pier Ferdinando Casini, Francesco Cossiga, Bettino Craxi, Massimo D'Alema, Alcide De Gasperi, Ciriaco De Mita, Antonio Di Pietro, Amintore Fanfani, Gianfranco Fini, Ugo La Malfa, Aldo Moro, Pietro Nenni, Achille Occhetto, Marco Pannella, Romano Prodi, Giuseppe Saragat, Giovanni Spadolini, Palmiro Togliatti, Valter Veltroni, Nichi Vendola.

There is also a corpus for the vocabulary that holds, for each word, the total number of occurrences, a Part-Of-Speech⁷ tag, its lemma, a count for the number of occurrences for each politicians, major party and opposition, and lastly the imprinting tag (e.g., masculine singular).

Lastly, **ParlaMint** [10, 11] is a compiled comparable parliamentary corpora for a number of countries and languages. ParlaMint corpora are interoperable, i.e. encoded to a very constrained common ParlaMint schema, a specialization of the Parla-CLARIN⁸ recommendations, which are a customization of the TEI Guidelines⁹.

It is a collection of parliamentary seating split into utterances. An example of two utterances could be *Politician*: "I ask permission to speak.", *President of the Senate*: "Permission granted.". For each utterance there are, among others, the following metadata: ID for both the whole seating and the utterance itself, a title of the seating, a date of the seating, the State Body that where it did take place (e.g., the Senate of the Republic), the name of the speaker and its affiliation, as well as the name of its coalition.

Speeches are collected before splitting in their own dataset, and for each speech we have, among others, a unique ID and a date, the type of speech (e.g., *Interview* or *Rally*), the venue channel (e.g., *BBC News*), the unique ID of the speaker and its role (e.g., *host* or *main speaker*), the text of the speech itself and its type (e.g., *Question*, *Answer*, *Interruption*) and whether there have been some *Incidents* and their type.

Incidents are breaks in the flow of the conversation, and they can be of different types¹⁰, for example "leaving" or "break". If an incident is of the kinesic type, it can be further subdivided into different kinesic types, such as "applause" or "laughter"¹¹. If the incident is of the vocal type, it can be further subdivided into different vocal types, such as "question" or "shouting"¹².

Speakers and Parties have also their own datasets, with unique IDs, names, affiliations, roles and the like.

Overall the taxonomy of ParlaMint is strongly codified, which make the interpretation of the dataset easier and machine-readable. All data is saved also in XML format.

4.1 My Own Corpora

Ultimately, I did not manage to find any corpus that I can directly integrate or use for my purposes, but I took inspiration from ParlaMint for the overall structure. Of course at the time of writing I still have not started compiling it and therefore the structure may be subjected of minor changes.

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⁷In grammar, a part of speech or part-of-speech (abbreviated as POS or PoS) is a category of words (or, more generally, of lexical items) that have similar grammatical properties [24].

⁸PrLa-CLARIN, a TEI schema for corpora of parliamentary proceedings: <https://clarin-eric.github.io/parla-clarin/>.

⁹P5 TEI Guidelines: <https://tei-c.org/guidelines/p5/>.

¹⁰Full list of incident types: "action", "incident", "leaving", "entering", "break", "pause", "sound", "editorial".

¹¹Full list of kinesic types: "kinesic", "applause", "ringing", "signal", "playback", "gesture", "smiling", "laughter", "snapping", "noise".

¹²Full list of vocal types: "greeting", "question", "clarification", "speaking", "interruption", "exclamat", "laughter", "shouting", "murmuring", "noise", "signal".

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